

VU Medical Zone

BIF 501 Midterm Fall 2021

VUTES 7.2 Fall 2021 (MidTer... x

BIF501 Bioinformatics II

Question No : 2 of 23 Marks: 1 (Budgeted)

Sequence in Genbank format is saved in chunks of Characters

Answer (Please select your correct option)

☐ 10

☐ 12

☐ 18

☐ 9

Answer solved By Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

Start Time: 8:56 AM

55:00
Time Left

Question Summary

<http://exam/vutes/client/Paper.aspx> store your password for exam? More info

Yes

DELL

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BIF501 Bioinformatics II

Question No : 11 of 23

MS2 genome has amino acids

Answer (Please select your correct option)

☐ 122

☐ 134

☐ 156

☐ 129

Answer solved By Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

Start Time 8:56 AM

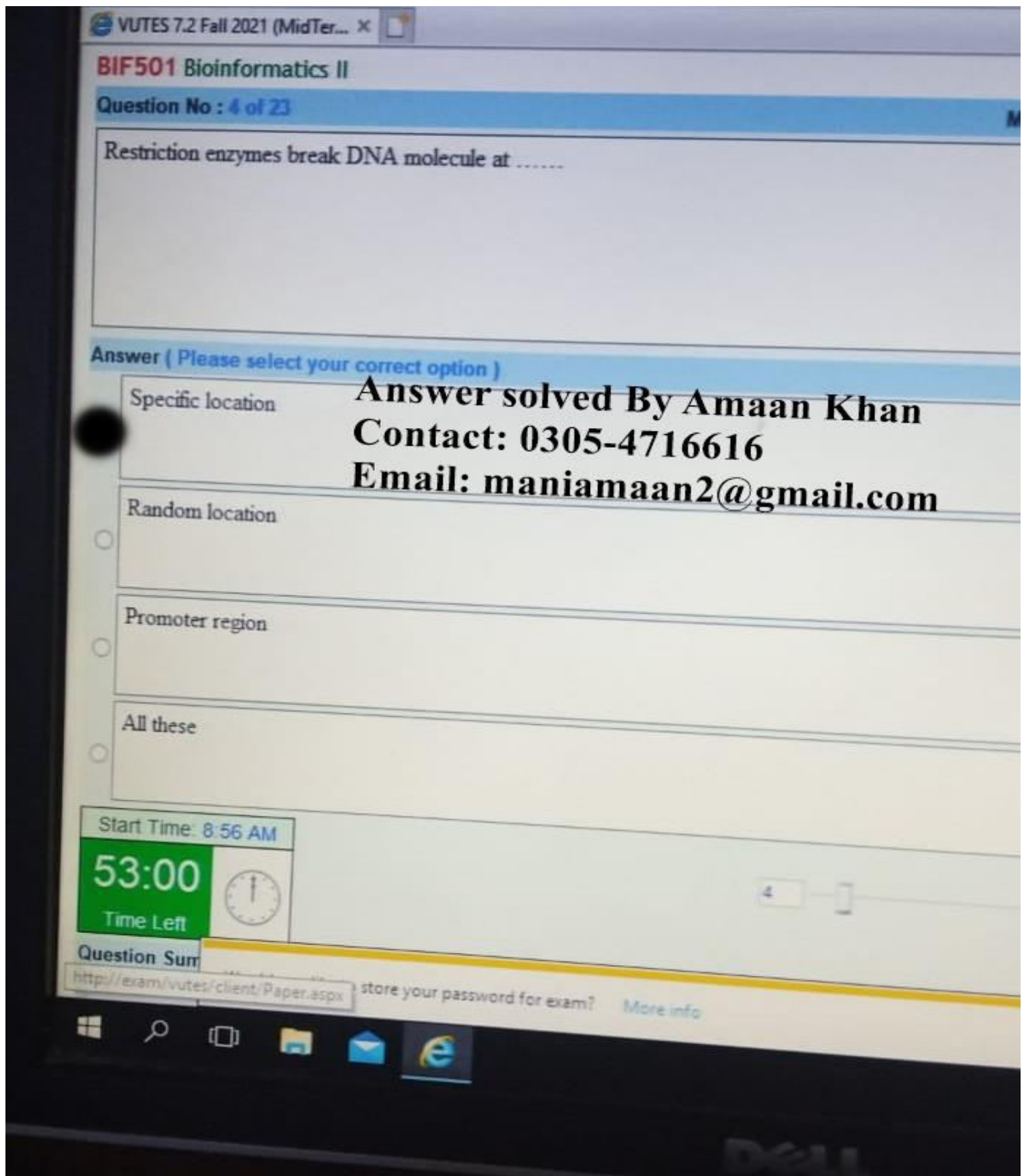
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Question Sum

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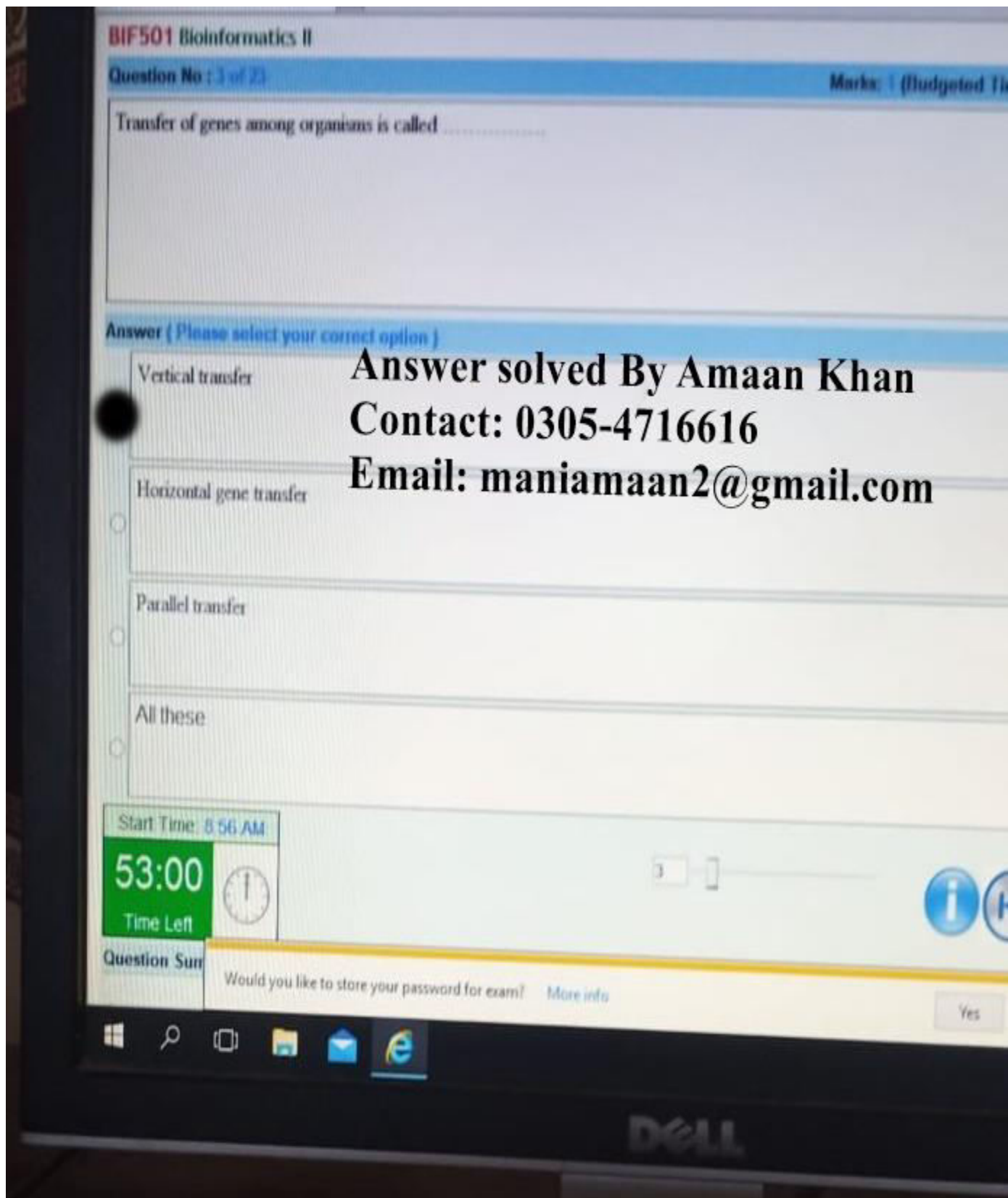
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BIF501 Bioinformatics II

Question No : 1 of 23

EMBL stands for

Answer (Please select your correct option)

- ☐ European molecular biotechnology lab
- ☐ European medium biology lab
- ☐ European molecular biology limitations
- ☐ European molecular biology lab

Answer solved By Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

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Question Sum

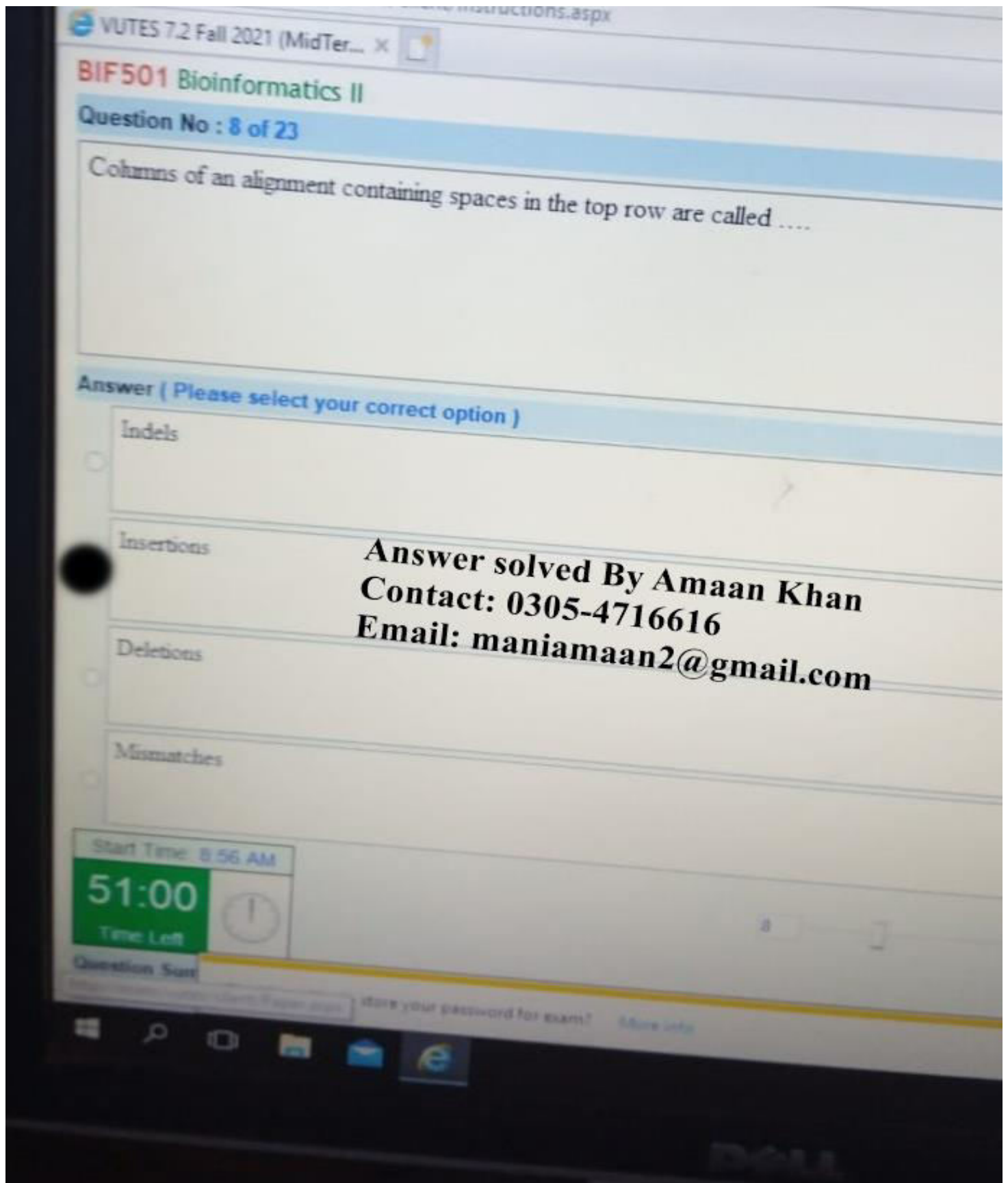
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BIF501 Bioinformatics II

Question No : 7 of 23

Marks:

Genome rearrangement results in change of gene

Answer (Please select your correct option)

Order

Duplication

Transfer

Transfusion

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Question Summary

Would you like to store your password for exam? [More info](#)

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BIF501 Bioinformatics II

Question No : 5 of 23

Marks: 1 (Bu

TCGGGGATTTC sequence in Drosophila helps in activation of gene

Answer (Please select your correct option)

☐ Cancer

☐ HCV

☐ HBV

☐ Immunity

Answer solved By Amaan Khan
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Email: maniamaan2@gmail.com

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Question Sum

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BIF501 Bioinformatics II

Question No : 12 of 23 Marks: 1 (B

What is the L value of a binary search tree having depth 5 starting from root.

Answer (Please select your correct option)

☐ 4

☐ 5

☐ 6

☐ 3

Answer solved By Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

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Question Summary : (Attempted Question ■)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

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BIF501 Bioinformatics II

Question No : 9 of 23 Marks: 1 (Budget

Symbols and search text of pattern are chosen from a predetermined finite set, called as

Answer (Please select your correct option)

☒ Alphabet

☐ Letters

☐ Numbers

☐ None of the above

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Question Sum

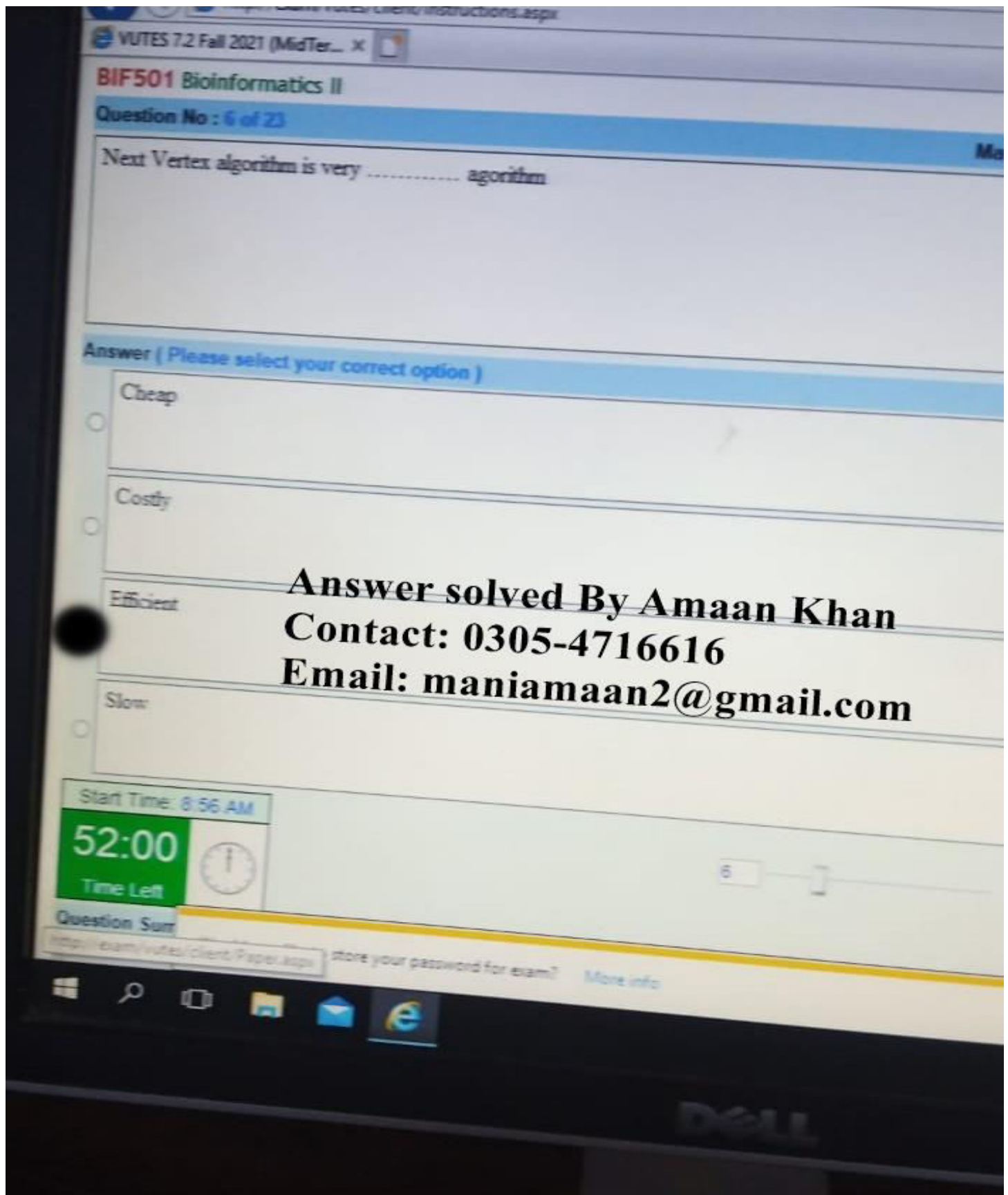
<http://exam.vutes/client/Paper.aspx> store your password for exam? More info Yes

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BIF501 Bioinformatics II

Question No : 10 of 23

KMP stands for

Answer (Please select your correct option)

☐ Knuth More Pratt

☐ Kane Micheal Pratt

☐ Knuth Morris Pratt

☐ None of the above

Answer solved By Amaan Khan
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Email: maniamaan2@gmail.com

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Question Sum

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BIF501 Bioinformatics II

Question No : 14 of 23

Marks: 1 (Budgeted Time)

Gene prediction algorithms for prokaryotes tend to be somewhat ___ than those for eukaryotes

Answer (Please select your correct option)

☐ Easier

☐ complex

☐ simpler

☐ difficult

Answer solved By Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23

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BIF501 Bioinformatics II

Question No : 15 of 23

An ORF is a sequence of DNA that starts with start codon

Answer (Please select your correct option)

☐ TAA

☐ TAG

☐ TGA

☒ ATG

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

Answer solved By Amaan Khan
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BIF501 Bioinformatics II

Question No : 8 of 23

Columns of an alignment containing spaces in the top row are called

Answer (Please select your correct option)

☐ Indels

☒ Insertions

☐ Deletions

☐ Mismatches

Answer solved By Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

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Question Summary : (Attempted Question)

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BIF501 Bioinformatics II

Question No : 16 of 23

Marks: 1 (Budgeted Time

DNA parts which can move from one part of the genome to other are called

Answer (Please select your correct option)

☒ Transposable elements

☐ Introns

☐ Exons

☐ Histones

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Question Summary : (Attempted Question)

2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23

Answer solved By Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

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Question No : 13 of 23 Marks: 1 (Budgeted)

_____ All mutations alter the functions of a protein

Answer (Please select your correct option)

☒ Not

☐ Yes

☐ About

☐ All Options are correct

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13

Question Summary : (Attempted Question)

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BIF501 Bioinformatics II

Question No : 20 of 23

Genome repeats
Define following terms
Tandem repeats
Interspersed repeats

Answer ([Please click here to Add Answer](#))

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Question Summary : (Attempted Question)

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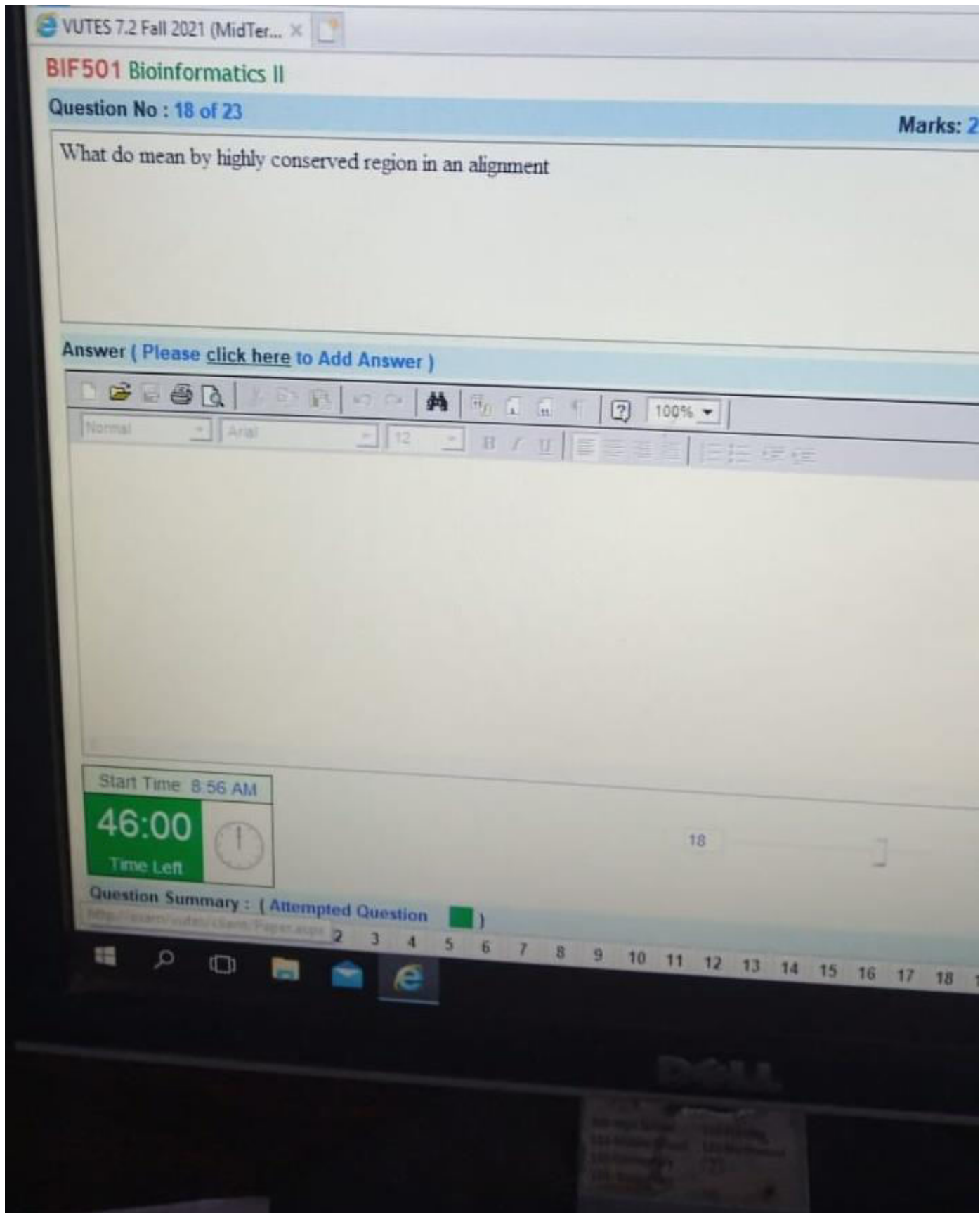
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A **tandem repeat** is a sequence of two or more DNA base pairs that is repeated in such a way that the repeats lie adjacent to each other on the chromosome.

Interspersed repeats are repeated sequence wherein the copies are interspersed, i.e dispersed throughout the genome. This makes it different from the tandem repeats wherein the copies lie side by side. In the interspersed repeats, the copies are nonadjacent.

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Introns start on the

5' ends as GT and

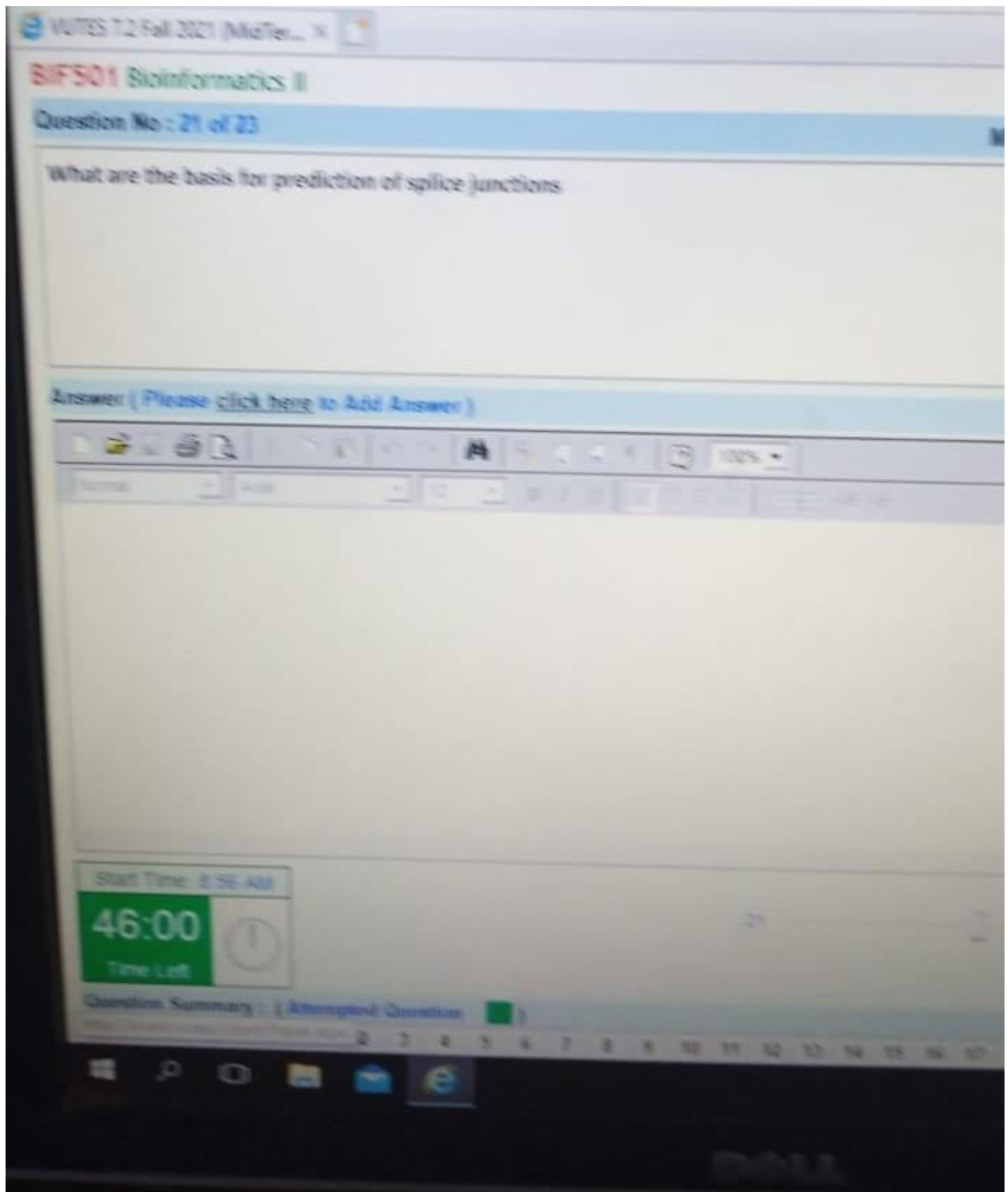
end as AG towards

3' end GT-----

-AG

**• This trend is highly
conserved**

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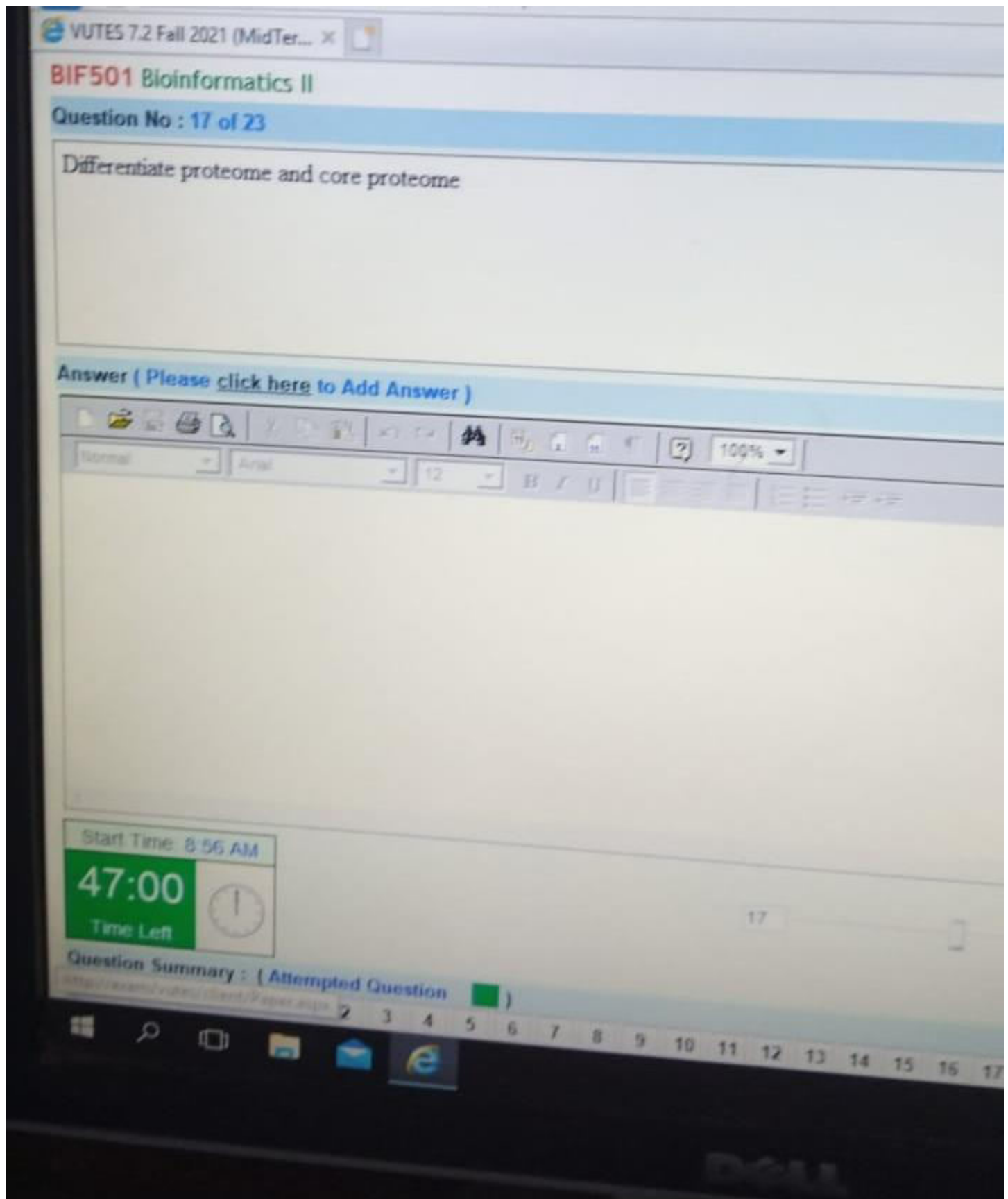


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Like TSS, the flanks of splice junctions show “biased” distributions of nucleotides in certain positions

- **These biased distributions of nucleotides are the basis for prediction of splice junctions**

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Core Proteome

- **All-against-all comparison provides an indication of Gene/Protein families**
- **Unique set of proteins is core proteome**

The proteome is the entire set of proteins that is, or can be, expressed by a genome, cell, tissue, or organism at a certain time. It is the set of expressed proteins in a given type of cell or organism, at a given time, under defined conditions.

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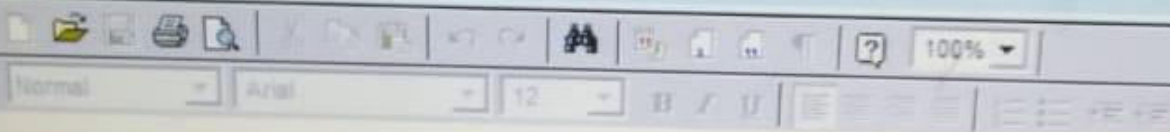
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SIF501 Bioinformatics II

Question No : 23 of 23

Describe pattern finding in biology and computational means?

Answer ([Please click here to Add Answer](#))



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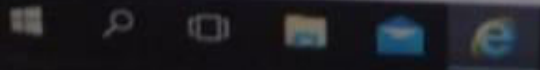
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23

Question Summary : (Attempted Question)

2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17



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Pattern Finding

In Biology it means finding motifs in DNA or proteins while in computational means it is finding a pattern in a string. After a genome is assembled, genome annotations are performed to identify gene and other features in a genome.

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BIF501 Bioinformatics II

Question No : 22 of 23

Explain Enzyme Commission Numbers by giving an example

Answer ([Please click here to Add Answer](#))

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22

Question Summary : (Attempted Question)

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Enzyme Commission (EC) numbers

Enzyme Commission (EC) under IUBMB

Enzymes are classified on the basis of reactions they catalyze

EC a.b.c.d a one of 6 classes of biochemical reactions b group of substrate c acceptor molecule d details of biochemical reaction

tripeptide aminopeptidases

EC 3.4.11.4

3 Hydrolases (use water to break substrate)

3.4 Hydrolases that act on peptide bonds